

PIZZA SALES ANALYTICS

Business Intelligence Project: Problem Statement



EXECUTIVE SUMMARY

This document outlines the analytical requirements for evaluating pizza sales performance. The project aims to compute key performance indicators (KPIs) and build a series of visualizations that together will form an interactive Power BI dashboard. The findings will support data-driven decision-making across sales, operations, and marketing functions.

PART 1: KPI REQUIREMENTS

The following five key performance indicators are required to provide a high-level summary of overall business performance from the pizza sales dataset:

1. Total Revenue	The aggregate sum of the total price across all pizza orders. This metric provides a direct measure of overall income generated within the reporting period.
2. Average Order Value (AOV)	The mean amount spent per order, derived by dividing Total Revenue by the Total Number of Orders. AOV indicates customer spending behaviour and helps benchmark promotional effectiveness.
3. Total Pizzas Sold	The cumulative sum of quantities across all order line items. This figure reflects overall product demand and volume throughput.
4. Total Orders	The distinct count of orders placed within the period. This KPI measures customer transaction frequency and overall order volume.
5. Average Pizzas Per Order	The mean number of pizzas included per order, calculated by dividing Total Pizzas Sold by Total Orders. This reveals typical basket size and can inform upselling strategies.

PART 2: CHARTS REQUIREMENTS

Seven visualizations have been identified to support trend analysis, category comparisons, and product-level performance evaluation. Each chart serves a specific analytical purpose within the dashboard:

1. Daily Trend for Total Orders

Chart Type: *Bar Chart*

A bar chart displaying the daily volume of orders across the reporting period. This visualization surfaces day-of-week or date-specific patterns, helping the business identify peak days, low-demand periods, and operational planning opportunities.

2. Monthly Trend for Total Orders

Chart Type: *Line Chart*

A line chart illustrating the hourly distribution of orders throughout the day. By mapping order activity against the clock, this chart identifies peak service hours and helps optimize staffing, kitchen capacity, and promotional timing.

3. Percentage of Sales by Pizza Category

Chart Type: *Pie Chart*

A pie chart showing each pizza category's share of total sales. This visualization reveals which categories (e.g. Classic, Veggie, Supreme, Chicken) are most popular and quantifies their respective contributions to overall revenue.

4. Percentage of Sales by Pizza Size

Chart Type: *Pie Chart*

A pie chart representing the proportion of sales attributed to each pizza size (S, M, L, XL, XXL). This chart uncovers customer size preferences and informs inventory and pricing decisions.

5. Total Pizzas Sold by Pizza Category

Chart Type: *Funnel Chart*

A funnel chart presenting total units sold for each pizza category in descending order. The funnel format makes it easy to compare category sales volumes at a glance and identify which categories drive the most quantity.

6. Top 5 Best Sellers by Revenue, Quantity & Orders

Chart Type: *Bar Chart*

A bar chart highlighting the five highest-performing pizza products ranked across three dimensions: Total Revenue, Total Quantity Sold, and Total Orders. This chart guides menu promotion and production prioritization for the most popular items.

7. Bottom 5 Best Sellers by Revenue, Quantity & Orders

Chart Type: *Bar Chart*

A bar chart showcasing the five lowest-performing pizza products across the same three dimensions. This chart is critical for identifying underperforming or unpopular offerings that may warrant menu review, repricing, or discontinuation.

